



SYMBIOSIS INTERNATIONAL (DEEMED UNIVERSITY)

(Established under section 3 of the UGC Act 1956)

Re - accredited by NAAC with 'A' Grade

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Founder: Prof. Dr. S. B. Mujumdar, M.Sc., Ph.D. (Awarded Padma Bhushan and Padma Shri by President of India)

Course Name: Agentic AI & Automation
Course Code:
Faculty: Engineering
Course Credit: 3
Course Level: 3
Sub-Committee (Specialization): Computer Science

Course Objectives:

- To introduce students to the concept of AI agents and the OpenAI Agents SDK for building single-agent and multi-agent systems.
- To enable students to design and coordinate multi-agent workflows with role assignment, handoffs, and guardrails.
- To develop practical skill in agentic frameworks such as AutoGen, LangGraph, CrewAI, and MCP.
- To build automation solutions using n8n with external services such as Gmail, Google Sheets, and Google Calendar.
- To apply Python-based data analysis and classical ML models inside agentic workflows for predictive and decision-support tasks.

Pre-requisites:

- Basic programming in Python.
- Familiarity with notebooks, APIs, JSON, and package installation.
- Introductory understanding of machine learning concepts.
- Very basic exposure to LLMs / prompt-based interaction.

Course Outline (Theory and Lab):

Unit. No.	Topic	Actual Teaching Hours	Contact Hours Equivalence
1	Introduction to AI Agent, Environment setup & Anaconda download for Mac, Windows, Linux, Build and run a first simple AI agent without tools or memory, and monitor its activity using traces on the OpenAI API platform, Add SQLite-based session storage to enable persistent memory and manage agent state, Apply memory-enabled agents to real use cases for multi-turn queries, Extend agents by integrating OpenAI tools (e.g., Code Interpreter) and combining real-time search, memory, and reasoning into a complete workflow, Leverage the OpenAI Agents SDK to build an AI agent with existing OpenAI tools such as code interpreter, Build AI Agents that can leverage external tools that integrate Tavily web search for structured, real-time search results, Wrap the search function as a FunctionTool for use within the OpenAI Agents SDK	9	9
2	AI Agent Team Structure, Manager Function Responsibilities, Design multi-agent pipelines for advertising campaigns, involving role-specific agents like Creative Director and Copywriter, Design a multi-agent research assistant that can	9	9

	triage user queries, delegate tasks to specialist agents, and generate an executive-ready report, Develop a library of specialist agents (e.g., Planner, Writer, Fundamentals Analyst, Search Agent) and coordinate their interactions, Implement guardrails for boundaries (e.g., politics), Build handoff mechanisms (e.g., Planner → Writer),Agent-as-Tool-Design		
3	Basic Concept: Multi-Model AI Agents in AutoGen, Building Multiple AI Agents with Different LLMs, Develop team of AI Agents for Marketing Strategy, Make the AI agents discuss ideas using AutoGen's chat, Multi-Model AI Agent in Autogen with Gemini & OpenAI's GPT-4o, Core Components of LangGraph, Agentic workflow in LangGraph, Configure API key, Connect to UI (Gradio), Visualize Agentic workflow, Define Custom Tool	9	9
4	Core components of CrewAI, Build Advanced Agentic AI Workflows Using CrewAI, Custom Tool: Python code (for data analysis and modelling), Train, linear regression, random forest, & XG-Boost models to predict sales using Scikit-Learn (Classical ML), Assemble a crew of AI Agents to perform the regression analysis to perform predictive analytics, Load, inspect, and understand the structure of the dataset (Classical ML), Handle missing values using appropriate imputation techniques, Perform Exploratory Data Analysis (EDA) and preprocess data, including cleaning and encoding categorical features, Evaluate and compare model performance using metrics like R ² , MAE, and MSE, Components of MCP, Transform Gradio Applications into an MCP-compatible tool by configuring Gradio parameters, Build an AI Agent using OpenAI Agents SDK with external MCP tools, Build and expose a new tool as an MCP service using Gradio, Deploy Gradio apps as MCP servers, Write Python code to act as an MCP client by discovering tool capabilities through manifests and remotely executing actions via HTTP requests, Simulate an AI assistant within the notebook that decides when and how to interact with external MCP tools.	9	9
5	Build Agentic Workflows Using n8n., Build a simple agent that takes user inputs, summarizes them using an AI model, and outputs the results, Integrate n8n with Google Sheets: Automate the process of sending AI-generated summaries and structured outputs into Google Sheets, Implement structured output parsing: Format AI responses as structured JSON, parse them within n8n, and store the results systematically, Build an agent capable of checking calendar events and creating new events based on user commands, Set up Gmail to trigger workflows that classify emails and automate further actions like calendar bookings, Combine Gmail triggers, email classification, Google Sheets tagging, & Google Calendar event creation into an automation process, Learn how to visually connect services like Gmail, Google Sheets, OpenAI, and your own APIs to automate tasks.	9	9
Total		45	45

Books Recommended:

- Hands-On Large Language Models — 2024 — Jay Alammar, Maarten Grootendorst — O'Reilly Media, Inc. (<https://www.oreilly.com/library/view/hands-on-large-language/9781098150952/>)
- Prompt Engineering for Generative AI — 2024 — James Phoenix, Mike Taylor — O'Reilly Media, Inc. (<https://www.oreilly.com/library/view/prompt-engineering-for/9781098153427/>)
- AI Engineering — 2024 — Chip Huyen — O'Reilly Media, Inc. (<https://www.oreilly.com/library/view/ai-engineering/9781098166298/>)
- Designing Machine Learning Systems — 2022 — Chip Huyen — O'Reilly Media, Inc. (<https://www.oreilly.com/library/view/designing-machine-learning/9781098107956/>)

Suggested Pedagogy:

- Short concept lectures followed by live demonstrations.
- Lab-based coding sessions for each framework and unit.
- Guided notebook exercises for single-agent and multi-agent workflows.
- Case-based learning using real automation scenarios.
- Mini-projects and a capstone-style integration task across tools and frameworks.

Evaluation/Assessment Methods:

- Quizzes on concepts and framework architecture.
- Laboratory assignments for each unit.
- Coding exercises on agents, tools, and automations.
- One or more mini-projects integrating multiple frameworks.
- Final project/demo with viva or presentation.

Course Outcomes (CO):

CO1: Implement single AI agents using the OpenAI Agents SDK with memory, tracing, and tool integration features for handling user tasks.

CO2: Develop multi-agent workflows with role-based agents, task delegation, guardrails, and handoff mechanisms using the OpenAI Agents SDK.

CO3: Build agentic applications using AutoGen and LangGraph frameworks with multi-model agents, custom tools, and workflow visualization.

CO4: Implement AI workflows using CrewAI and MCP by integrating predictive analytics, external tools, and MCP-compatible services.

CO5: Create automation workflows in n8n by integrating AI services with Gmail, Google Sheets, and Google Calendar for task automation and structured data handling.

Expert (Internal/External):

Dr. Shreyas R. Hole
SIT, Nagpur

Benchmarking:

- CS294/194-196 Large Language Model Agents — University of California, Berkeley — <https://rdi.berkeley.edu/llm-agents/f24>
- 15-482 Autonomous Agents — Carnegie Mellon University —

- **Agentic AI Strategies and Applications — Indiana University Kelley School of Business** — <https://kelley.iu.edu/executive-education/professional-development/ai-and-tech/ai-agents/index.html>
- **AI Agents PG Program for Business Applications — The University of Texas at Austin, McCombs School of Business** — <https://onlineexeced.mcombs.utexas.edu/ai-agents-for-business-applications-online-course>

Select Sustainable Development Goal (SDG) of the Course: (mention in the next column)	Answer (mention the SDG of the course)
☐ No poverty (SDG 1) ☐ Zero hunger (SDG 2) ☐ Good health and well-being (SDG 3) ☐ Quality education (SDG 4) ☐ Gender equality (SDG 5) ☐ Clean water and sanitation (SDG 6) ☐ Affordable and clean energy (SDG 7) ☐ Decent work and economic growth (SDG 8) ☐ Industry, innovation and infrastructure (SDG 9) ☐ Reduced inequalities (SDG 10) ☐ Sustainable cities and communities (SDG 11) ☐ Responsible consumption and production (SDG 12) ☐ Climate action (SDG 13) ☐ Life below water (SDG 14) ☐ Life on land (SDG 15) ☐ Peace, justice, and strong institutions (SDG 16) ☐ Partnerships for the goals (SDG 17)	SDG 4: Quality Education SDG 8: Decent Work and Economic Growth SDG 9: Industry, Innovation and Infrastructure